

HPC - Get to know your GPU

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1 Implementation

The experiment was run in the GPU of Google Colab notebook.

```
import numba
from numba import cuda

cuda.detect()

# Device name
device = cuda.select_device(0)

print(f"Device id: {device.id}")
print(f"Device name: {device.name}")

# Core info
multiprocessor_count = device.MULTIPROCESSOR_COUNT
# Look up CUDA cores per SM from dict based on Compute Capability (64 for
# Compute Capability 7.5)
cores_per_SM = 64
total_cores = multiprocessor_count * cores_per_SM

print(f"Multiprocessor count: {multiprocessor_count}")
print(f"Total cores: {total_cores}")

# Memory info
mem_info = cuda.current_context().get_memory_info()
print(f"Total memory: {mem_info[1]}")
```

2 Results

The GPU information obtained using `numba.cuda` is summarized in Table 1.

Table 1: GPU information reported by `numba.cuda` on Google Colab.

Property	Value
Device ID	0
Device name	Tesla T4
Compute capability	7.5
Multiprocessor count	40
Cores per SM	64
Total CUDA cores	2,560
Total memory	15,637,086,208 bytes (14.56 GiB)